

Day 13: DAX Filter Context & CALCULATE() Mastery

[DEF] Definition to Remember

CALCULATE() is the single most powerful function in DAX. It is the only function that can modify, override, expand, or clear the existing Filter Context of a visual.

1. The Two Evaluation Contexts in DAX

1. Row Context (Iterative): Knowing the values of the *current single row*. Present in calculated columns and iterator functions (SUMX).

2. Fil

2. Anatomy of `CALCULATE()`

```
CALCULATE(
    <Expression / Measure>,
    <Filter_Modifier_1>,
    <Filter_Modifier_2>,
    ...
)
```

3. Modifying Filter Context Examples

```
// Example 1: Overriding Category Filter
Audio Product Sales =
CALCULATE(
    [Total Sales],
    Dim_Product[Category] = "Audio"
)

// Example 2: Multiple Filter Conditions (AND logic)
US High Value Sales =
CALCULATE(
    [Total Sales],
    Dim_Customer[Country] = "United States",
    Fact_Sales[Quantity] > 5
)
```

4. Context Transition

When a DAX measure is invoked inside a Row Context (like inside a calculated column or SUMX), DAX automatically initiates Context Transition, transforming the current row's key into an active Filter Context.

[MUST-WRITE] Must-Master Skills & Interview Points

1. CALCULATE() evaluates all filter arguments *in parallel* first, then evaluates the main expression under the newly modified filter context.
2. Explain Context Transition: how invoking a measure turns a row context into a filter context.
3. Why CALCULATE is the core building block of all advanced BI metrics.

[RECALL] Quick Recall

CALCULATE = Modifies Active Filter Context | Context Transition = Row Context -> Filter Context